STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service	**)**	**Case 25-E-0072**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service	**)**	**Case 25-G-0073**

DIRECT TESTIMONY

OF

DAVID LYON

ON BEHALF OF

ENVIRONMENTAL DEFENSE FUND

Dated: May 30, 2025

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I. Introduction and Qualifications

Q: Please state your name, title, and business address.

A: My name is David Lyon. My title is Senior Methane Scientist II at Environmental Defense Fund. My business address is 2060 Broadway Street, Suite 300, Boulder, CO 80302.

Q: On whose behalf are you submitting this testimony in this proceeding?

A: I am submitting this testimony on behalf of Environmental Defense Fund ("EDF").

Q: Please provide a summary of your education and experience.

A: I earned a PhD in Environmental Dynamics from the University of Arkansas in 2016 with a dissertation on methane super-emitters from the oil and gas industry, a Master of Science in Forestry from the University of Kentucky in 2004, and a Bachelor of Arts in Biology from Hendrix College in 2002. I have over 15 years of experience working on oil and gas methane emissions, greenhouse gas emissions accounting, and emissions inventories, while at the Arkansas Department of Environmental Quality, Environmental Defense Fund, United States Environmental Protection Agency, and University of Texas at Austin. I have co-authored over 30 peer-reviewed papers on oil and gas methane emissions, including *Assessment of methane emissions from the US oil and gas supply chain* (Alvarez et al. 2018), which has been cited over 1000 times. To see more information, please see my attached CV, Exhibit __ (DL-2), or visit https://scholar.google.com/citations?hl=en&user=Y32HsP0AAAAJ.

Q: Have you previously provided testimony before regulatory or legislative bodies?

A: Yes. I have testified several times, including before the United States House of Representatives Committee on Science, Space, and Technology in 2022 for the hearing *Detecting and Quantifying Methane Emissions from the Oil and Gas Sector*; and before the U.S. Environmental Protection Agency in 2022 for the International Emissions Inventory Conference. In 2018, I was a member of the New York State Energy Research and Development Authority technical advisory committee for developing an oil and gas emissions inventory.

II. Purpose of Testimony and Recommendations

Q: What is the purpose of your testimony?

A: The purpose of my testimony is to (1) evaluate the greenhouse gas emissions data presented by Consolidated Edison Company of New York, Inc. ("Company" or "Con Edison") in its gas and electric rate filings; (2) present a more comprehensive assessment of greenhouse gas emissions associated with the Company's proposed rate plan, that will enable the Commission and stakeholders to evaluate the rate plan's consistency with the state emissions limits established under the Climate Leadership and Community Protection Act ("CLCPA"); and (3) present recommendations for improved GHG emissions accounting in this and future rate cases, to enable assessments under the CLCPA.

Q: Please provide a summary of your testimony and recommendations.

A: New York State must reduce statewide greenhouse gas ("GHG") emissions by 40% by 2030 and 85% by 2050, below 1990 levels, and achieve economywide net zero emissions

by 2050, to fulfill the directives of the CLCPA.[1] State agencies, including the New York Public Service Commission ("NYPSC" or "Commission"), must consider whether their decisions will interfere with the attainment of the statewide GHG emissions limits—and if so, the agency must adopt mitigation measures to address the related emissions. *See* CLCPA § 7(2). The Commission has sought to evaluate its decisions in this manner, including in utility rate cases, but it has not deployed numeric analysis to verify the overall GHG emissions impact of a given utility rate plan. To accurately assess whether approval of a utility rate plan will interfere with the attainment of New York's GHG emissions limits, a comprehensive quantification of the emissions associated with the rate plan is required.

Con Edison included some assessment of GHG emissions in its rate filing, but the Company's emissions analysis is incomplete.

Using the best available information, and using emissions accounting practices consistent with New York's climate law, EDF presents an analysis of the Company's electric and gas system emissions over the term of the rate filing—2026, 2027, and 2028. The emissions are also compared against the emissions-reduction trajectory that was modeled in the New York Climate Action Council's Scoping Plan to achieve the statewide 2030 emissions limit. The analysis demonstrates that the Company's rate filing is not consistent with the GHG emissions reductions needed to achieve the statewide

[1] New York State Climate Leadership and Community Protection Act, 2019 N.Y. Laws 106 § 2, available at https://legislation.nysenate.gov/pdf/bills/2019/S6599 [hereinafter "CLCPA"]; N.Y. ECL §§ 75-0103(11), 75-0107(1).

2030 or 2050 emissions limits. See Exhibit __ (DL-1) for detailed analysis and methodology.

Finally, this testimony provides recommendations to improve GHG emissions accounting in this and future rate cases. Here, EDF sought to obtain relevant information from Con Edison through discovery. In the future, Con Edison and the Commission should standardize providing granular GHG emissions information in a consistent format.

Please note that all figures and tables referenced in testimony are included in Exhibit __ (DL-1). This testimony does not seek to formulate any legal interpretations; any legal statements are provided on advice of counsel.

Q: Are you providing any exhibits to your testimony?

A: Yes. I am attaching the following exhibits to my testimony:

- **Exhibit __ (DL-1):** EDF Assessment of Con Edison NY 2025 Rate Case GHG Emissions
- **Exhibit __ (DL-2):** Curriculum Vitae
- **Exhibit __ (DL-3):** 2025 CECONY N. Gas & Electric Rate Cases- GHG Impact (Overall)
- **Exhibit __ (DL-4),** Con Edison Response to NRDC-2-10 (May 2, 2025)
- **Exhibit __ (DL-5),** Con Edison Response to EDF-2-29 (May 19, 2025) (including Attachment 1)
- **Exhibit __ (DL-6),** Con Edison Response to EDF-2-30 (May 19, 2025)
- **Exhibit __ (DL-7),** Con Edison Response to EDF-3-32 (May 28, 2025)
- **Exhibit __ (DL-8),** Con Edison Response to EDF-3-34 (May 28, 2025)
- **Exhibit __ (DL-9),** Con Edison Response to EDF-1-16 (May 2, 2025)

III. Quantification of the GHG Emissions Associated with Con Edison's Rate Filing is Needed to Evaluate its Climate Impact

Q. Why is it urgent to reduce GHG emissions?

A. Scientific evidence overwhelmingly demonstrates that climate change is causing immediate, devasting impacts, and that these harms will worsen dramatically as greenhouse gas pollution continues to rise.[2] The Intergovernmental Panel on Climate Change ("IPCC") stated that "[h]uman influence has warmed the climate at a rate that is unprecedented in at least the last 2000 years."[3] Cutting GHG emissions now is critical because "there is a near-linear relationship" between human-caused GHG emissions and related global warming, meaning that each additional increment of global warming exacerbates changes in extreme weather events.[4] Preventing the worst impacts of climate change "requires steep and sustained reductions in fossil fuel production and use," in addition to measures that reduce production-cycle emissions.[5]

Q. Why is it important to reduce methane emissions?

A. Methane, the primary component of natural gas, is itself a potent greenhouse gas; and when combusted, methane further contributes to warming by releasing the GHG carbon dioxide (CO_2). Compared with CO_2, methane's warming potential is ~84 times higher

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2 *See* Crimmins et al, *Fifth National Climate Assessment*, U.S. GLOB. CHANGE RSCH. PROGRAM, Ch. 1, at 5 (Emily K. Laidlaw et al. eds) (2023), https://nca2023.globalchange.gov/downloads/NCA5_2023_FullReport.pdf.

3 INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE ("IPCC"), *Climate Change 2021: The Physical Science Basis: Summary for Policymakers* (2021), at 6, https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_SPM.pdf.

4 IPCC, *Climate Change 2021: The Physical Science Basis: Summary for Policymakers* (2021), at 28.

5 SEI, IISD, ODI, E3G, & UNEP, THE PRODUCTION GAP REPORT, *Summary of Findings,* at 1, https://productiongap.org/2021report/.

over a twenty-year period and ~30 times higher over a 100-year period.[6] Given the urgent global need to reduce greenhouse gas emissions to mitigate the climate crisis, cutting methane emissions is an important strategy to limit the rate of current warming. Peer-reviewed research demonstrates that rapid implementation of all methane mitigation measures could slow the rate of near-term warming by 30 percent and avoid a quarter of a degree centigrade of additional warming by midcentury.[7]

Q. Are there clear pathways to reduce methane emissions from the oil and gas supply chain?

A. Yes. Best available data suggest that methane emissions from the oil and gas sector account for 20 percent or more of total global emissions.[8] As measurement capabilities improve, there is also evidence that methane emissions from this sector have been historically underestimated. For example, the measurement-based study Alvarez et al. 2018 estimates that the former U.S. national inventory underestimated actual methane emissions from the oil and gas sector by as much as 60 percent.[9]

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6 IPCC AR6, WG 1, The Physical Science Basis at 7-125.

7 Ocko et al., *Acting rapidly to deploy readily available methane mitigation measures by sector can immediately slow global warming*, 16 ENV. RESEARCH LETTERS 054042 (2021), https://iopscience.iop.org/article/10.1088/1748-9326/abf9c8.

8 Schwietzke et al., *Aerially guided leak detection and repair: A pilot field study for evaluating the potential of methane emission detection and cost-effectiveness*, 69 J. AIR & WASTE MGMT. ASSOC. 71 (2019), https://doi.org/10.1080/10962247.2018.1515123; Hmiel et al., *Preindustrial 14CH4 indicates greater anthropogenic fossil CH4 emissions*, 578 NATURE 409 (2020), https://doi.org/10.1038/s41586-020-1991-8; Saunois et al., *The Global Methane Budget 2000–2017*, 12 EARTH SYSTEM SCIENCE DATA 1561 (2020), https://doi.org/10.5194/essd-12-1561-2020.

9 Alvarez et al., *Assessment of methane emissions from the U.S. oil and gas supply chain*, 361 SCIENCE 186 (2018), https://doi.org/10.1126/science.aar7204.

Yet many mitigation measures exist already and could significantly limit methane emissions from human activities. According to the International Energy Agency (IEA), the U.S. could abate nearly 40 percent of methane emissions from the oil and gas sector at no net cost, and currently available technologies could abate up to 76 percent of oil and gas methane emissions.[10]

Q. How is New York State addressing climate change?

A. New York State has recognized the existential threat of climate change and the need for rapid climate mitigation and adaptation. The CLCPA directs the state to reduce statewide GHG emissions 40% by 2030 and 85% by 2050 from 1990 levels and to achieve economy-wide net zero emissions by 2050 (the remaining 15% reductions can come from carbon offsets).[11] The CLCPA seeks to ensure that GHG emissions reductions are "real, permanent, quantifiable, verifiable, and enforceable," and to "encourage early action to reduce [GHG] emissions and co-pollutants."[12] And the climate law specifically acknowledges the short-term climate impact of methane emissions, recognizing that methane causes 84 times as much global warming as the same amount of carbon dioxide.[13]

Q. How is New York State addressing climate change in relation to the power sector?

A. The CLCPA sets a detailed trajectory to ensure a clean power sector, directing the Commission to set standards requiring that 70% of statewide electric generation secured

[10] IEA, *Methane Tracker* (last updated May 7, 2025), https://www.iea.org/articles/methane-tracker-data-explorer.

[11] CLCPA § 2; N.Y. ECL §§ 75-0103(11), 75-0107(1).

[12] CLCPA § 2; N.Y. ECL §§ 75-0109(3)(b), (d).

[13] CLCPA § 2; N.Y. ECL § 75-0101(2).

by jurisdictional load-serving entities be generated by renewable energy by 2030, and that the state's power sector system be zero emissions by 2040.[14]

Q. Are Con Edison's GHG emissions relevant to achieving the statewide emissions limits in 2030 and 2050?

A. Yes. Con Edison provides gas and electric service to over 10 million people, with approximately 3.6 million electric customers and 1.1 million gas customers across New York City and Westchester.[15] The Company delivers approximately 53,000 gigawatt hours ("Gwh") of electricity to customers annually, and projects continuing these delivery volumes, with slight growth, during 2026-2028.[16] The Company delivers approximately 168 million dekatherms of natural gas to customers annually, and projects continuing these delivery volumes with some decrease during 2026-2028.[17] There are numerous sources of GHG emissions associated with the Company's provision of gas and electric service, including the upstream emissions associated with oil and gas production and transportation, the emissions associated with delivery and end-use combustion of natural gas (aka methane), and the emissions associated with out-of-state and in-state power generation and transmission for electric customers.

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[14] CLCPA § 4; N.Y. PSL § 66-p.

[15] *See* Ex. __ (CETP-1) at 5; CON EDISON, *Our Businesses*, https://www.conedison.com/en/about-us/our-businesses (last accessed May 26, 2025).

[16] *See Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service,* Case 25-E-0072, Direct Testimony of Electric Forecasting Panel at p6 (Jan. 31, 2025).

[17] *See Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service,* Case 25-G-0073, Direct Testimony of Gas Forecasting Panel at p5-6 (Jan. 31, 2025).

Q. What obligations apply in this rate case relating to the achievement of the statewide GHG emissions limits?

A. There are numerous obligations created by New York's climate law that affect Con Edison's operations and the programs and investments proposed in this rate case. Here, we focus on a particular obligation: State agencies, including the NYPSC, must consider whether their decisions will interfere with the attainment of the statewide GHG emissions limits—and if so, the agency must adopt mitigation measures to address the related emissions.[18] Logically, in order to make such a determination, clear quantification of the GHG emissions associated with a given proposal under review—and subsequent Commission decision—is required.

Q. How has the Commission acted to ensure fulfillment of this obligation?

A. First, the Commission has specifically found that the obligation to assess consistency with statewide emissions limits applies to its decisions in utility rate cases.[19] Second, the Commission has established guidelines for utilities to collect needed information: In May 2022, the Commission identified the need to "establish clear and consistent statewide guidelines for GHG emissions reporting requirements to ensure that the State's major electric and gas IOUs (collectively, the Utilities) are on track to meet the CLCPA

[18] *See* CLCPA § 7(2).

[19] *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service & KeySpan Gas East Corp. d/b/a National Grid for Gas Service*, Cases 19-G-0309 & 19-G-0310, Order Approving Joint Proposal, As Modified, & Imposing Additional Requirements at 69 (Aug. 12, 2021).

targets."[20] The Commission stated that it "directs all Utilities in future rate filings to include an assessment of the GHG emissions impacts of each specific investment, capital expenditure, program, and initiative included in their rate filings."[21]

Q. What more is needed for the Company, stakeholders, and the Commission to assess the GHG emissions associated with the rate filing?

A. To accurately assess whether approval of a utility rate plan will interfere (or not) with the attainment of New York's GHG emissions limits, a comprehensive quantification of the emissions associated with the rate plan is required. The Commission has directed utilities to include in their rate filings "an assessment of the GHG emissions impacts of each specific investment, capital expenditure, program, and initiative," but more precise implementation of this directive is required. First, utilities must actually provide this level of granularity in their rate filings, and the emissions assessment should cover all the years of the rate plan. Second, because rate cases in New York are often resolved through settlements that differ from the utility's original rate filing, such settlements—or Joint Proposals—should contain an updated GHG emissions assessment, enabling the Commission to evaluate them under the CLCPA. Third, the Commission should include a quantitative assessment of the GHG emissions impact in its order on the rate case, not merely a qualitative discussion as it has done in the past.

[20] *In the Matter of Assessing Implementation of and Compliance with the Requirements and Targets of the Climate Leadership and Community Protection Act*, Case 22-M-0149, Order on Implementation of the CLCPA at 14-15 (May 12, 2022).

[21] *Id.* at 16.

IV. GHG Data Provided in Con Edison's Gas & Electric Rate Filings

Q. Did Con Edison provide a GHG emissions assessment within its rate filing?

A. The Company provided some analysis of GHG emissions in its rate filing, but the assessment is incomplete. Con Edison states that "consistent with recent PSC Orders, the Company provides projections of the greenhouse gas ('GHG') emissions reductions the proposals presented in these proceedings are anticipated to achieve."[22] The Company included several documents in its gas and electric rate filings that quantify GHG emissions associated with the Company's gas and electric businesses, summarized below:

i. **Direct Testimony of the Clean Energy Transition Panel** ("CETP") – Discusses Con Edison's approach to the clean energy transition, and how the proposals in these rate cases help the company move towards achievement of the CLCPA's GHG emission limits. The testimony focuses primarily on Con Edison's strategy for decarbonization pursuant to the CLCPA and other relevant laws, including those enacted by the City of New York. The testimony provides a narrative discussion of GHG emissions quantification at Page 71 Line 24, through Page 74 Line 14.
   a. The CETP states that Con Edison's "direct emissions from the natural gas distribution system and electric system SF6 emissions" are approximately 660,000 MT CO2e, and refers to Exhibit __ (CETP-2), Schedule 1 for more detail on "direct GHG impacts." Page 73 Lines 4-7, 18-20.
   b. The CETP states that Con Edison's "proposed projects and programs result in an estimated net reduction in GHG emissions of over 39,000 MT CO2e." Page 73 Lines 10-12.
   c. The CETP states that Con Edison's "anticipated indirect GHG emissions reductions are approximately 7 million MT CO2e and are set forth in Exhibit ___ (CETP-2), Schedule 2." Page 74 Lines 12-14.

ii. **Exhibit __ (CETP-1), Con Edison Integrated Long Range Plan (2025)** ("ILRP") – Presents the Company's planning strategy to reduce emissions and to achieve the state's decarbonization goals.

[22] Cases 25-E-0072 & 25-G-0073, Direct Testimony of Clean Energy Transition Panel at 16, lines 18-21 (Jan. 31, 2025).

a. Consistent with the Company's Gas System Long-Term Plan (2023),[23] the ILRP focuses on two potential pathways—identified as Hybrid and Deep Electrification—by which Con Edison could fulfill the requirements of the CLCPA and other climate policies. The first part of the report is a narrative of the approaches, programs, and overarching strategy for the company's energy transition plan; the second part is a series of appendices that give a high-level view of the assumptions and characteristics of the two decarbonization pathways.
b. In the ILRP, however, the Company does not commit to either the Hybrid or Deep Electrification pathway. This approach is, again, consistent with the Company's approach in the Gas System Long-Term Plan proceeding—where the Company stated in its Final Long-Term Plan that "[t]he Reference Pathway will continue to inform our short-term rate case funding requests."[24] The Commission stated in its order on the Plan that: "the Companies chose not to select a preferred Pathway in their Final LTP"; "[t]he Commission disagrees with this approach"; and found that "for purposes of clarification, the Commission understands that the Companies have selected the Reference Pathway as their Final LTP."[25]
c. Because the Commission and the Company have stated that they are following the Reference Pathway, EDF relies on this pathway in its GHG emissions analysis.

iii. **Exhibit __ (CETP-2-UPD)** – includes the following tables:
a. **Schedule 1: 2025 N. Gas & Electric Rate Case- GHG Impact Summary** – Identifies "direct" emissions reductions from the natural gas distribution system, Con Edison's vehicle fleet, and Con Edison facilities.
b. **Schedule 2**:
i. **Indirect Carbon Emissions (GHG) Reductions (2026 through 2028)** – Identifies "indirect" emissions reductions associated with building electrification and the Company's efforts to support the electric infrastructure necessary for the expansion of electric vehicle charging.
ii. **Clean Energy Investments and Programs: Indirect Carbon Emissions (GHG) Calculations** – Identifies the inputs used to calculate the prior table.

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[23] *In the Matter of a Review of the Long-Term Gas System Plans of Consolidated Edison Co. of N.Y., Inc. & Orange & Rockland Utilities, Inc.*, Case 23-G-0147, Con Edison & O&R Utilities, Final Gas System Long-Term Plan (Nov. 29, 2023).

[24] *Id. at* 5 (Nov. 29, 2023).

[25] Case 23-G-0147, Order Regarding Long-Term Natural Gas Plan & Requiring Further Actions at p29-30 (Sept. 20, 2024).

iv. **Exhibit __ (CETP-3), Disadvantaged Communities Report** – Fulfilling a commitment by Con Edison in the 2023 Joint Proposal to track and report on how its operations affect disadvantaged communities.

Q. Please provide the primary tables included in the Company's GHG emissions assessment of its rate case filing.

A. Below are the two primary tables provided in Con Edison's GHG emissions assessment of its rate case filing at Exhibit __ (CETP-2-UPD), with formatting adjusted for ease of presentation. See these and the third table at Ex. __ (DL-1) at p2-3.

Figure A. Source: Con Edison Ex. __ (CETP-2), Schedule 1.

2025 N. Gas & Electric Rate Case- GHG Impact Summary					
	2023	**2026**	**2027**	**2028**	**Total Projects Savings (MT CO2e)**
N. Gas Distribution System	623,047	-10,374	-10,444	-10,487	-31,305
Electric	See Schedule 2				
Vehicles	29,393	-1,498	-1,327	-771	-3,596
CECONY Facilities*	27,110	-922	0	0	-4,367
Total Emissions Reductions					**-39,268**
*Proposed CECONY Facilities projects will not all be completed within the Rate Years (2026, 2027, and 2028). Phase 1 projects will be completed by the end of 2026 and Phase 2 projects are expected to be completed by the end of 2029. The Total Projects Savings for CECONY Facilities shows the total projected savings from all proposed projects upon completion.					

Figure B. Source: Con Edison Ex. __ (CETP-2-UPD), Schedule 2, p1.

Indirect Carbon Emissions (GHG) Reductions (2026 through 2028)						
Category	**Source of Reduction**	**Program**	**Tons of CO2e Reduction**			
			2026	**2027**	**2028**	**Total**
Building Electrification (BE)	Electrification	Space/Water Heating, Cooking, and Clothes Drying	670,709	1,024,662	1,417,385	3,112,757
Electric Vehicles (EV)	Electric Vehicle Adoption	Vehicle Programs and Non-utility program adoption	789,851	1,136,183	1,547,141	3,473,175
Total			1,556,946	2,300,387	3,151,593	**6,585,932**

For further discussion of the information provided by the Company, see Ex. __ (DL-1), EDF Assessment of Con Edison NY 2025 Rate Case GHG Emissions, p2-5.

Q. Please identify concerns you have with the Company's approach to emissions accounting.

A. The Company's effort to provide a quantification of GHG emissions associated with its gas and electric rate filings is appreciated. However, there are a number of gaps and concerns with the submission:

- **<u>Lack of framework for emissions data that would include total annual emissions and detailed emissions breakdown</u>** – The Company presents GHG emissions information in a confusing and vague manner. Tables are dispersed, not clearly labeled, and lack sufficient detail. Emissions should be presented within a framework that includes total annual emissions by scope and business, detailed breakdown within scopes, identification of rate case-related reductions within that framework, and

provide, for comparison and assessment, an CLCPA-compliant emissions trajectory for the scope and business.

  - The Schedule 1 table (Fig. A above; Ex. __ (DL-1) Fig. 2), for instance, apparently shows Con Edison's "direct" emissions, but it only provides those for the Gas System, not the Electric System, and it provides no breakdown into subcategories. The Schedule 2 tables (see Fig. B above; Ex. __ (DL-1) Figs. 3, 4) apparently show Con Edison's "indirect" emissions, but they do not provide estimates of the total indirect emissions for the Gas or Electric System, showing only the projected emissions *reductions* for each rate year.
  - To assess the extent of any projected reductions, a baseline is required. The Company should show the projected GHG emissions profile for each year of the rate plan, with a breakdown of inputs, alongside the projected emissions reductions. This should be organized by Scope 1, 2, and 3 emissions, and should include a complete accounting of all Scope 1, 2, and 3 emissions.

- **Lack of clear emissions baseline** –There is no emissions baseline presented for the Electric System—only the projected effect on electricity demand shown in Exhibit __ (CETP-2-UPD), Schedule 2. Exhibit __ (CETP-2), Schedule 1 provides a 2023 emissions baseline for the Gas System, but insufficient detail is provided: The CETP testimony states that the Company relied on the December 1, 2022, GHG Emissions Inventory Proposal (as supplemented by the JU's May 31, 2023 filing), filed in Case 22-M-0149; but tables showing these calculations should be presented. The highest-level calculations were made available as workpapers in response to a Department of Public Service Staff information request, attached as Exhibit __ (DL-3). The

Company should provide a clear GHG emissions baseline for both Gas and Electric Systems, including Scope 1, 2, and 3 emissions.

- **Lack of clear methodology** – In the absence of a uniform methodology for GHG emissions accounting, the Company should explain its methodology and sources for such calculations. For example, in Schedule 2, the Company provides the sources for its inputs regarding contributors to emissions *reductions* – building and vehicle electrification – but it does not provide the same level of detail for calculating the total emissions.
- **Assessment of Exhibit ___ (CETP-2), Schedule 1** –
  - **Missing electric system emissions** – The only electric systems emissions quantified by the Company are the "Indirect" emissions reductions associated with facilitating building electrification and uptake of electric vehicles, *see* Exhibit __ (CETP-2-UPD), Schedule 2. The Company should also quantify the GHG emissions associated with electricity delivered to customers — although most power generation is not in the Company's direct control, it is still relevant to the analysis.
  - **"Vehicles" category should be clarified** – Schedule 1 should make clear that the "Vehicles" line appears to refer only to Company-operated vehicles; not the broader electric vehicle initiative addressed in Schedule 2.
- **Assessment of Exhibit ___ (CETP-2-UPD), Schedule 2** – The calculations appear to be simplified for estimation (which is acceptable), the emission factors used in the calculations look plausible, and the source data is clearly identified at Schedule 2 p2—though we were unable to locate all of the sources cited. For building

electrification, some data is lacking that would help more fully understand the program, particularly the assumptions regarding quantity of residences and commercial building addressed over the three years. In the updated version, which includes additional factors for electrification, it is not clearly defined what the assumed adoption rates of the electrified equipment are. For the vehicle electrification data, the "2025 CECONY Rate Case EV Fleet Forecast" was not included in the rate filing, which would be valuable context to better understand this program.

Q. Did Con Edison provide additional data or information regarding its GHG emissions?

A. Yes, the Company provided additional information in response to information requests from EDF and other parties to the rate case. That information is utilized in EDF's analysis presented below. However, the Company should be providing sufficient information in its initial rate filing to understand its GHG emissions assessment, without extensive discovery being required. For further discussion, see Exhibit __ (DL-1) at p2-5. Part VI of this testimony, at page 31, presents detailed recommendations to improve future GHG emissions accounting in rate cases.

V. An Assessment of GHG Emissions Associated with Con Edison's Gas & Electric Rate Filings

Q. What statewide information and assumptions did EDF consider in its assessment of Con Edison's GHG emissions?

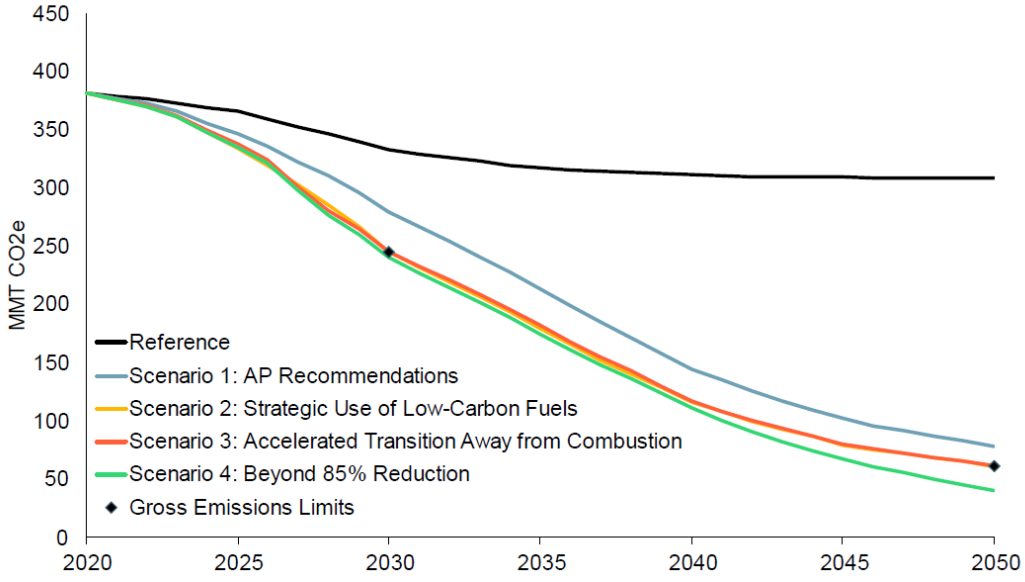
A. State agencies, including the NYPSC, must consider whether their decisions are inconsistent with, or will interfere with, the attainment of the statewide GHG emissions limits. The Commission has not set GHG emissions limits or guidelines for individual

utilities or for utilities by type; nor have other state agencies. For purposes of the Con Edison gas and electric rate cases, we interpret "consistency" to mean that the company would achieve GHG emission levels in the years covered by the rate case (2026, 2027, and 2028) that are on a trajectory consistent with achievement of the statewide emissions limit, as applied to the Company's gas and electric businesses. This analysis particularly focuses on the 2030 statewide emissions limit as it is closest in time.

The New York State Climate Action Council's Scoping Plan provides a range of scenarios designed to meet the CLCPA goals, shown below.

Figure C. Source: Scoping Plan Integration Analysis, Sec. 1 p19 (Dec. 2022).

Figure 12. GHG Emissions by Mitigation Scenario



Under these scenarios, the decline in consumption of fossil natural gas by 2030 ranges between 34% for Scenario 3 (Accelerated Transition Away from Combustion) and 37%

for Scenario 4 (Beyond 85% Reductions).[26] Electricity demand by 2030 increases between 7% for Scenario 2 (Strategic Use of Low Carbon Fuels) and 12% for Scenarios 3 and 4, while emissions from electricity decline between 67% for Scenarios 3 and 4 and 70% for Scenario 2. For the purposes of this methodological review, we have selected Scenario 3 as a benchmark to represent a trajectory that aligns with the CLCPA.

Q. What company-specific information did EDF consider in its assessment of Con Edison's GHG emissions?

A. Because, as stated above, the Company did not provide estimates of systemwide gas or electric emissions for the years of the rate plan, EDF used available information to calculate such estimates. In this rate case, Con Edison asserts that its actions are consistent with the Reference Case presented in the Company's 2025 Integrated Long Range Plan, Exhibit __ (CETP-1). For example, the Company stated that "[t]he capital investments proposed and described in the Company's filings are wholly based on the Reference Case," as opposed to the "Deep Electrification" and "Hybrid" scenarios modeled in its ILRP and identified as compliant with CLCPA goals. *See* Exhibit __ (DL-4), Con Edison Response to NRDC-2-10 (May 2, 2025). Thus, EDF looked to the GHG emissions projected by the Company in its Reference Case analysis in Exhibit __ (CETP-1), the 2025 ILRP.

[26] New York State Climate Action Council Scoping Plan (Dec. 2022), Integration Analysis Technical Supplement, Section I, Annex 2: Key Drivers and Outputs, available at https://climate.ny.gov/Resources/Scoping-Plan. *See* worksheet Final Energy Demand by Fuel ("All_Energy_by_Fuel"); note that the fossil natural gas consumption excludes use for "electricity generation, synthetic fuel production, or electrolysis."

Figure D., from Con Edison ILRP.

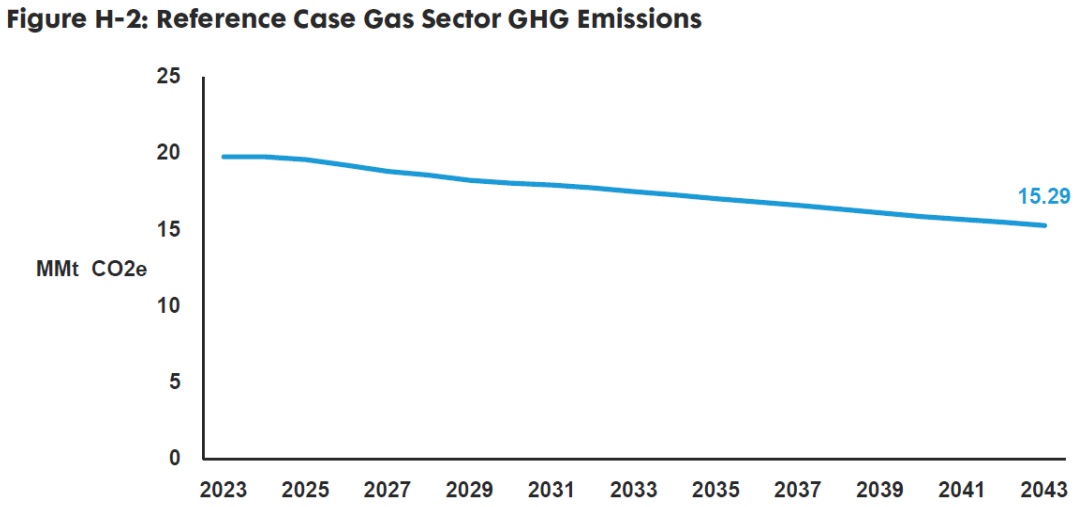
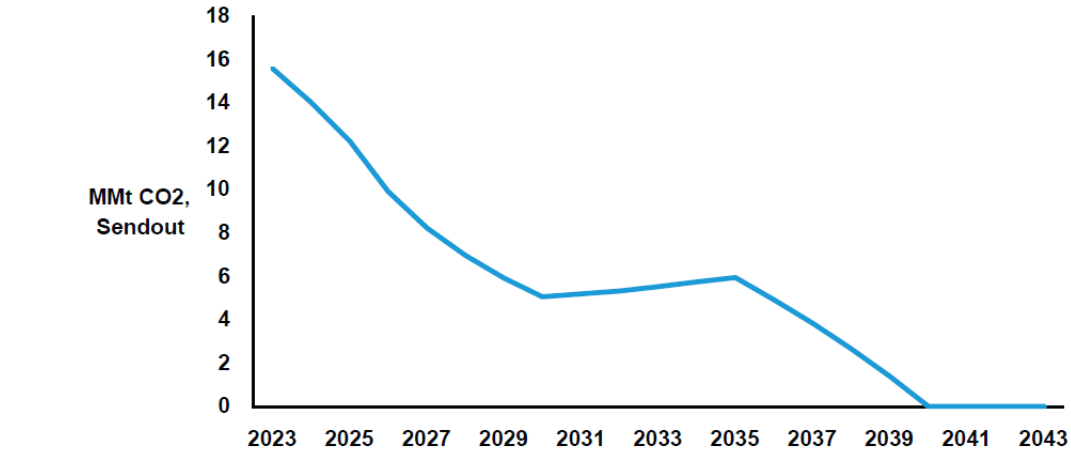


Figure E., from Con Edison ILRP.



The company has additionally, upon request, provided partial underlying data for Exhibit __ (CETP-1), ILRP Figures H-2 and H-3 (for 2023-2028), and for ILRP Figures A-2 (Reference Gas Customer Sales) and A-4 (Reference Electric Volume Forecast). *See* Exhibit __ (DL-5), Con Edison Response to EDF-2-29 (May 19, 2025); Exhibit __ (DL-9), Con Edison Response to EDF-1-16 (May 2, 2025); Exhibit __ (DL-4), Con Edison Response to NRDC-2-10.

A. Gas System Emissions

Q. Please present an assessment of Con Edison's gas system GHG emissions.

A. See Figures F and G below; see Exhibit __ (DL-1) for a more detailed explanation of methodology. This assessment compares the following key elements:

- Red: Estimated GHG emissions for the forecasted years of the rate case (2026-2028)
- Blue: Long-term GHG emissions trajectory of Con Edison's gas system under the Company's Integrated Long-Range Plan Reference Case
- Green: Long-term GHG emissions trajectory of Con Edison's gas system under a CLCPA-compliant trajectory from the Scoping Plan

Figure F.

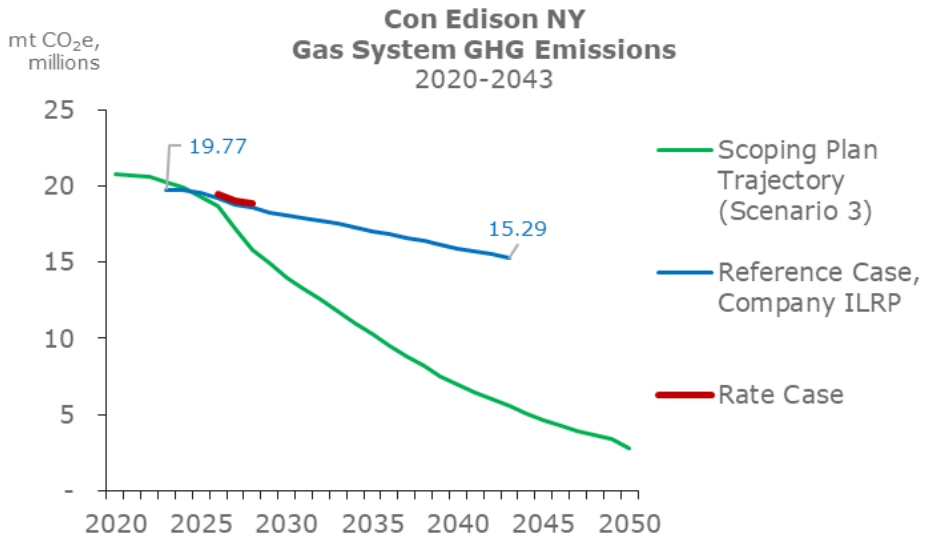
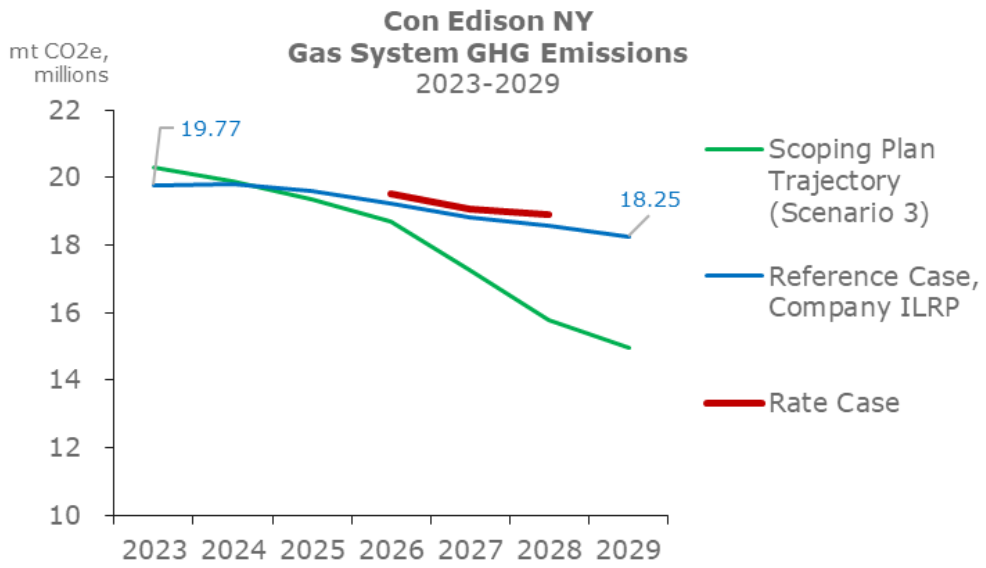


Figure G.



Q. What is shown by the gas system emissions analysis?

A. The gas system GHG emissions associated with Con Edison's rate case filing do not appear to be in accordance with the Scoping Plan, and are not aligned with the GHG emissions reductions needed by 2030 to align with the CLCPA-mandated statewide emissions limit. The gas system rate case emissions even appear to be slightly higher than the Company's own Reference Case from its ILRP.

Q. What is the basis of the analysis?

A. The analysis draws from the New York State Scoping Plan, the Company's Long-Term Gas System Plan, Integrated Long-Range Plan, rate case filings, and responses to information requests. A detailed explanation of the methodology supporting this analysis is presented in the attached Exhibit __ (DL-1) at p5-10.

Q. Are there any concerns or limitations with the analysis?

A. Yes. Con Edison chooses to presume a lower GHG emissions intensity for differentiated natural gas, which the Company refers to as certified natural gas. The Company assumes that "Certified" natural gas is responsible for just 22,215 gCO2e/mmbtu compared with 41,671 gCO2e/mmbtu for "Not Certified" natural gas. *See* Con Edison CETP Direct, Exhibit _ (CETP-1) at p75 Table H-1; *see also* Exhibit __ (DL-1) at p8, Figure 9. EDF believes that the "certified" terminology and assumptions of reduced methane intensity—which suggests that improved performance standards have been implemented and verified—is premature given the variability and subjectivity of existing programs. Differentiated gas products are still nascent and lack safeguards to ensure credibility. The primary concerns are:

1. *Lack of Measurement, reporting and verification (MRV) standards:* Without comprehensive direct measurement, independent verification, and transparency around intensity calculations, it is inappropriate for producers or certifiers to label gas as responsibly sourced or to claim that gas is achieving a certain methane intensity.
2. *Limited participation:* The voluntary nature of certification programs limits operator participation and, in turn, opportunities for emission reductions. This may also lead to a lack of available certified gas for domestic consumers if operators prioritize certified gas for export.
3. *Limited coverage of participating companies:* Participating companies may choose to only certify gas from facilities that already have good emissions performance. This

would imply minimal or even zero additional emission reductions from certification programs.[27]

Q. What further context is relevant regarding "differentiated" or "certified" natural gas?

A. Con Edison is currently in the midst of a pilot project regarding certified gas acquisition, which was approved by the Commission as part of the 2023 Joint Proposal in the Company's last rate case.[28] EDF stated at the time that the program was "appropriately structured as a cost-contained pilot project with an objective of gathering information that can inform the company, the Commission, and the public about the status of differentiated gas markets and the ability of operators and certifiers to demonstrate that market differentiation is warranted in a transparent fashion."[29] In approving the Joint Proposal, the Commission characterized the Company's certified gas pilot as the type of "[p]rograms that test potential avenues to reduce GHG emissions"—acknowledging that the value of certified gas is uncertain—and concluded that a pilot program is appropriate to determine if certified gas "can be used to reduce GHG emissions during the transition period to a decarbonized gas system."[30] This context demonstrates that it is not

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[27] *See* Lackner et al., *Certification of Natural Gas with Low Methane Emissions: Criteria for Credible Certification Programs*, EDF (May 2022), https://blogs.edf.org/energyexchange/wp-content/blogs.dir/38/files/2022/05/EDF_Certification_White-Paper.pdf.

[28] *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric & Gas Service*, Cases 22-E-0064 & 22-G-0065, Joint Proposal at 93 (Feb. 16, 2023).

[29] Cases 22-E-0064 & 22-G-0065, EDF Statement of Neutrality on Joint Proposal at 9-10 (Mar. 29, 2023).

[30] Cases 22-E-0064 & 22-G-0065, Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans with Additional Requirements at 134 (July 20, 2023).

appropriate at this time for the Company to incorporate assumptions about reduced methane emissions associated with "certified" gas purchases into its GHG emissions accounting practices.

B. Electric System Emissions

Q. Please present an assessment of Con Edison's electric system GHG emissions.

A. See Figures H and I below. This assessment compares the following key elements:

- Red, Solid: Rate case emissions, using Rate Filing Emission Factors ("EFs") – Estimated GHG emissions associated with the Con Edison electric system for the forecasted years of the rate case (2026-2028), calculated based on the Company's Rate Case Filing, at Exhibit __ (CETP-2-UPD), Sched 2, p2.
- Red, Dotted: Rate case emissions, using Updated Reference Case EFs – Estimated GHG emissions associated with the Con Edison electric system for the forecasted years of the rate case (2026-2028), calculated based on an updated version of the Company's ILRP Reference Case, as disclosed at Exhibit __ (DL-5), Con Edison Response to EDF-2-29.
- Blue: Long-term GHG emissions trajectory of Con Edison's electric system under the Company's ILRP Reference Case, as updated and disclosed in Exhibit __ (DL-5), Con Edison Response to EDF-2-29.
- Green: Long-term GHG emissions trajectory of Con Edison's electric system under a CLCPA-compliant trajectory from the Scoping Plan (Scenario 3).

Figure H.

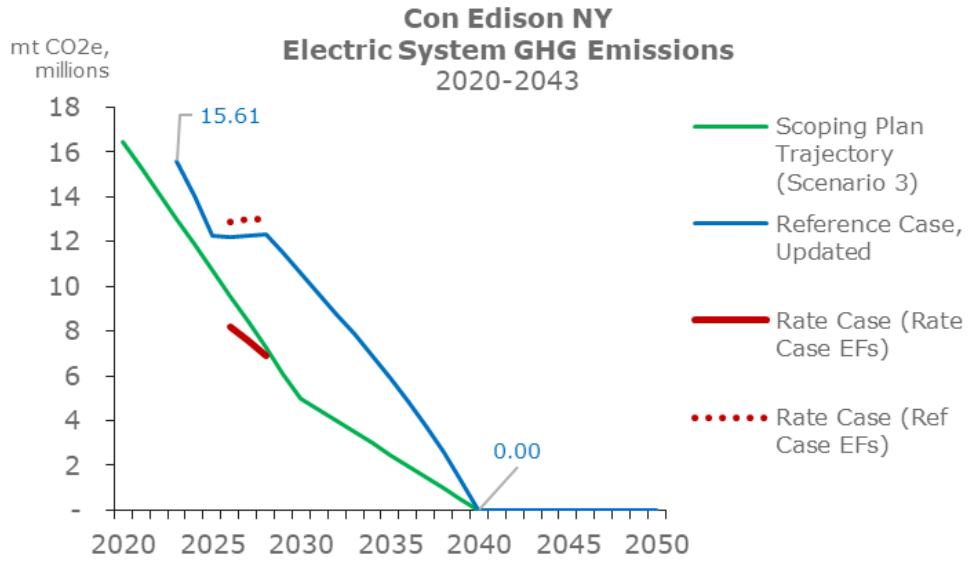
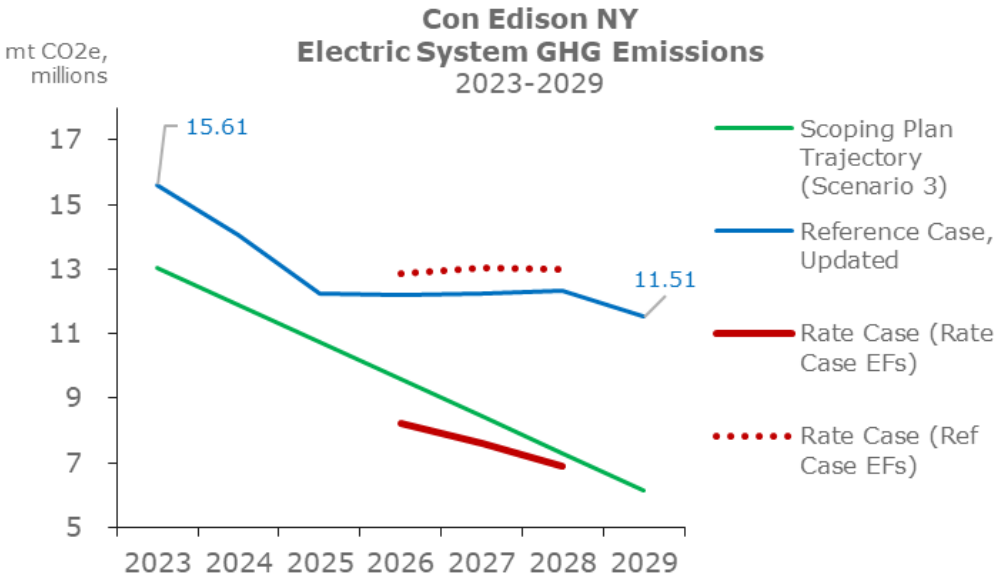


Figure I.



Q. What is shown by the electric system emissions analysis?

A. The analysis of electric system GHG emissions associated with Con Edison's rate case filing present uncertainty, since there are two sets of emissions factors ("EFs") provided by the Company, enabling two different quantifications of the GHG emissions associated with the electric rate case filing. One estimate appears to demonstrate that the electric system is exceeding the GHG reductions required consistent with the Scoping Plan (Rate Case EFs), while the other estimate appears to demonstrate that the system is not on track and is even exceeding the GHG emissions projected in the Company's ILRP (Updated Reference Case EFs). That the uncertainty in this instance lies with the estimate of future grid emission factors, which are not necessarily under Con Edison's control, highlights the sensitivity of the emission projections to those factors, and the need for clarity and consistency in their estimation and use.

Q. What is the basis of the analysis?

A. The analysis draws from the New York State Scoping Plan and the Company's Integrated Long-Range Plan, rate case filings, and responses to information requests. A detailed explanation of the methodology supporting this analysis is presented in the attached Exhibit __ (DL-1) at p10-15.

C. Claimed Emissions Reductions

Q. What GHG emissions reductions does Con Edison claim in its rate filing, and how do you evaluate those claims?

A. In Exhibit __ (CETP-2), Schedule 2, Con Edison identifies GHG emissions reductions associated with building electrification and electric vehicle adoption. Building electrification can contribute to reductions in the Company's Scope 3 emissions where

the Company is transitioning customers off of its fossil gas distribution system to instead purchase cleaner electricity to meet those energy needs. This is because Scope 3 emissions include those emissions associated with customer use of products sold by the Company—in this instance, reducing customer combustion of natural gas sold by Con Edison reduces the GHG emissions associated with that natural gas. Meanwhile, Con Edison's facilitation of electric vehicle adoption does not reduce the Company's Scope 3 emissions, since Con Edison is not selling gasoline to fuel internal combustion vehicles or otherwise supplying fossil fuel products as opposed to EVs. However, the Company does implement important programs that facilitate electric vehicle uptake, which is essential to achieve New York's statewide GHG emissions reduction targets—so it is appropriate for Con Edison to identify these reductions. They should be classified as a more distant emissions reduction contribution, though—not within Scope 1, 2, or 3 emissions. For more detailed discussion of emissions categorization, see *supra* at Part VI at p28.

Q. What is your assessment of the Company's building electrification emissions reduction quantification?

A. In Exhibit __ (CETP-2), Schedule 2, Con Edison identifies that building electrification is projected to reduce gas load by 7.5, 11.4, and 15.5 million mmBtu in the years 2026, 2027, and 2028 respectively (shown in Figure 3), and increase electric load by 0.8, 1.1, and 1.6 TWh, respectively in those same years. The claimed CO_2e emissions reduction shows a net reduction of 0.7, 1.0, and 1.4 million metric tons in 2026, 2027, and 2028, respectively. *See* Exh.__ (CETP-2), Schedule 2, p2; *see also* Ex.__ (DL-1) at p3, Fig. 4.

The net emissions shown are consistent with the electric and gas emission factors used in this exhibit for those years. The amounts of gas load reduced presented here, however, do not correspond to the identified load reductions in the Company's Gas Forecasting Panel exhibits which show reductions of 0.8, 0.7, and 0.9 million mmBtu, which would yield far lower emissions reductions than claimed in the exhibit. [31] (By comparison, the annual load increases are reflected in the Electric Forecasting Panel exactly as shown.)

Thus, it appears that the building electrification programs proposed in the rate case would serve a mixture of Con Edison electric customers, including some that are not currently Con Edison gas customers—such as customers who might use other heating fuels, or who customers who would switch off of gas service but are not in Con Edison's gas service territory.

Facilitating building electrification is essential to achieve New York's statewide GHG emissions reduction targets—so it is appropriate for Con Edison to identify these reductions. For customers not moving off of Con Edison's gas system, however, the GHG reductions achieved should be classified as a more distant emissions reduction contribution —not within Scope 1, 2, or 3 emissions.

VI. Improved GHG Accounting in Rate Cases

Q. Why is it important for Con Edison to report GHG emissions in Scope 1, 2, and 3 format?

[31] Con Edison Gas Forecasting Panel Direct, Ex. __ (GFP-6), pp 1-3. In this exhibit the reductions are shown in MDts of 781, 686, and 883.

A. Con Edison currently reports its GHG emissions in an aggregated format, without distinguishing between Scope 1, Scope 2, and Scope 3 emissions. While the Company does distinguish between "direct" and "indirect" emissions in some places, that approach is less precise to accurately categorize emissions. This lack of disaggregation presents a barrier to evaluating the utility's alignment with the Climate Leadership and Community Protection Act (CLCPA), which mandates aggressive, sector-specific emissions reductions and requires utilities to demonstrate measurable progress toward statewide decarbonization targets.

According to the Greenhouse Gas Protocol,[32] emissions are categorized as follows:

- Scope 1: Direct emissions from sources owned or controlled by the company.
- Scope 2: Indirect emissions, such as from the generation of purchased electricity, steam, heating, or cooling consumed by the company.
- Scope 3: All other indirect emissions that occur across the company's value chain, including upstream activities (e.g., purchased goods and services, capital goods) and downstream activities (e.g., customer use of sold products, such as natural gas).

Without scope-level granularity, it is difficult to determine which emissions sources are being addressed by proposed investments or operational changes, and whether those reductions are material in the context of Con Edison's total emissions footprint. For instance, a rate case may propose investments in electrification or grid modernization, which could reduce Scope 1 emissions (e.g., from gas infrastructure) or Scope 2

[32] GREENHOUSE GAS PROTOCOL, *How are scope 3 emissions organized? – Overview of the scopes*, https://ghgprotocol.org/scope-3-frequently-asked-questions-0#question%20one (last accessed May 27, 2025).

emissions (e.g., through cleaner purchased electricity). However, if Scope 3 emissions—such as those from customer use of natural gas—remain unaccounted for, the full climate impact of the utility's operations and its alignment with CLCPA goals cannot be accurately assessed. Moreover, aggregated reporting obscures the emissions intensity of specific business segments, making it difficult for regulators and stakeholders to evaluate the cost-effectiveness and climate benefit of proposed rate increases.

Q. What recommendations would you provide to improve GHG emissions accounting for Con Edison in this rate case and in future matters?

A. There are three buckets of recommendations to improve emissions accounting.

1. **<u>Establish a Clear Framework for Assessing Emissions</u>**: Utilities should have clarity about what they will be measured against to show success, and what the framework is by which to report emissions.
    a. Identify, prior to the assessment of a case's consistency with CLCPA goals, a clear trajectory of reductions by which a case would be measured. In the analysis above, we used the Scoping Plan Scenario 3 as a stand-in trajectory – the Commission could continue to use the Scoping Plan or use that as a starting place to adopt a more specialized trajectory. When GHG emission reductions are presented as part of a case, they should be mapped to defined CLCPA trajectories so that they can be readily assessed quantitatively as to their consistency with CLCPA goals. For the cases discussed here, that would include a trajectory for:
        - A Con Edison gas-specific emissions path,
        - A Con Edison electric-specific emissions path,

- External avoided emissions: paths by which the Commission could evaluate the Company's contribution to the electrification of vehicles and buildings that are not part of the Company's own footprint but that constitute. One consideration is whether/how these avoided emissions should be claimed and accounted for; and on the gas or electric side of the business. contributions to the State's achievement of the CLCPA emissions limits. *See* Ex. __ (DL-1) at 18.

b. The emissions factors and forecast basis of emission projections should be consistently defined and used in the evaluation.

2. **Disaggregate and Report GHG Emissions by Scope**: Con Edison, in this and future rate cases, should report the projected annual emissions for each proposed rate year, plus one year before and after the proposed rate years for context, for its gas & electric systems. This reporting should clearly distinguish between Scopes 1, 2, and 3 emissions in accordance with the GHG Protocol (see below), to enable more precise climate impact assessments.

3. **Attribute Emissions Reductions to Specific Proposals**: Emissions reductions should be (1) disaggregated and identified by their Scope consistent with the baseline projection described above, and (2) clearly linked to individual programs or investments proposed in the rate case, allowing stakeholders to evaluate their effectiveness and alignment with CLCPA targets. The Company should also report and quantify notable reductions that are not within the Company's Scopes 1, 2, and 3, such as facilitating vehicle electrification.

For more detailed Scope 1, 2, and 3 reporting recommendations, see Exhibit __ (DL-1) at p18-20.

These recommendations are consistent with the Commission's direction that "all Utilities in future rate filings [are] to include an assessment of the GHG emissions impacts of each specific investment, capital expenditure, program, and initiative included in their rate filings."[33]

Q. Does this conclude your testimony?

A. Yes.

[33] *In the Matter of Assessing Implementation of and Compliance with the Requirements and Targets of the Climate Leadership and Community Protection Act*, Case 22-M-0149, Order on Implementation of the CLCPA at 16 (May 12, 2022).